

LUIS RENTERIA LEZANO

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SUMMARY

Artificial Intelligence and Machine Learning graduate student (4.18/4.2 GPA) with an engineering background in industrial instrumentation and control. Builds and ships end-to-end machine learning systems in Python - training, inference and deployment - as containerized FastAPI microservices with CI/CD. Experience spans graph neural networks, retrieval-augmented generation, and AI-assisted (vibe) development, plus four years of engineering across refinery automation, IP infrastructure and enterprise networking.

TECHNICAL SKILLS

Languages: Python, SQL; HTML, CSS, JavaScript (foundational)

Machine Learning and Deep Learning: PyTorch, scikit-learn, XGBoost, PyTorch Geometric, Graph Neural Networks, Hugging Face Transformers, Optuna, SHAP, spaCy

GenAI and Retrieval: Retrieval-augmented generation (RAG), embeddings and vector search / HNSW, local LLMs, prompt engineering, AI agents, on-device AI

AI-assisted development: Vibe coding with AI developer tools (agentic workflows); shipping products end to end in unfamiliar stacks

MLOps and Tooling: Docker, FastAPI (async microservices), GitHub Actions (CI/CD), Git, Linux, Streamlit, Pytest, model monitoring

Data and Visualization: pandas, NumPy, SciPy, Power BI, Excel, Matplotlib, Seaborn

Industrial Data: Allen-Bradley PLC (Studio 5000), FactoryTalk View (HMI/SCADA), HART, Modbus, Profibus, Ethernet/IP, field instrumentation and calibration

PROJECTS

Cairn - On-Device RAG Study Assistant | EmbeddingGemma, ObjectBox HNSW, local LLM, Android | Capstone, 2026

- Designing the chunking and retrieval layer of a fully offline Android assistant that answers questions over user documents (PDF, DOCX, PPTX, TXT) with no network calls.
- Sentence-aware chunking sized to the embedding token limit; dense retrieval over an ObjectBox HNSW vector index using EmbeddingGemma, scoped per notebook to keep answers grounded.

Wisp - Cross-Platform Mobile App (shipped) | Expo, React Native, TypeScript, AI-assisted build | github.com/renteria-luis/wisp

- Designed the product and built it end to end via vibe coding - directing AI coding tools to implement it in a stack I had not used before - shipped with CI, strict type-checking and tests.
- A deterministic, fully-tested rule engine (never an LLM at runtime) computes each user's adaptive plan; 100% on-device with SQLite storage, local notifications and English/Spanish support.

Fraud Detection Pipeline | XGBoost, Optuna, FastAPI, Docker, GitHub Actions | github.com/renteria-luis/fraud-detection-v1

- Built an end-to-end fraud detection system on PaySim (349:1 class imbalance), benchmarking Logistic Regression, Random Forest and XGBoost; achieved 0.91 PR-AUC and 85% recall.
- Selected an operational threshold of 0.2226 over the 0.5 default to recover otherwise-missed fraud; deployed as an async FastAPI service with a multi-stage Docker build and CI on Hugging Face Spaces.

Graph-Based AML Detection | PyTorch Geometric, GraphSAGE, FastAPI | github.com/renteria-luis/graph-aml-detector

- Building a money-laundering detector on the Elliptic Bitcoin transaction graph (204K nodes) using GraphSAGE with neighbor sampling; chose it over GCN/GAT under an 8GB memory constraint.
- Temporal split over random to prevent leakage, with an XGBoost baseline to demonstrate the value the graph structure adds; a LangGraph ReAct agent layer is planned for explainable risk reports.

House Prices (Kaggle) and Telco Churn | scikit-learn, ensembles, boosting | github.com/renteria-luis

- House Prices: cross-validated feature engineering and a soft-voting ensemble (Lasso, RF, Gradient Boosting, SVR) - ranked #202 on the Kaggle public leaderboard. Telco Churn: soft-voting ensemble with threshold tuning to prioritize recall.

PROFESSIONAL EXPERIENCE

Applications Engineer | Bermalar Consulting S.A.C. | Lima, Peru | **February 2022 - March 2023**

Engineering Intern, September 2021 - February 2022. Three-engineer automation consultancy subcontracted to Cobra on the Talara

Refinery Modernization program (Petroperu).

- Programmed Allen-Bradley PLC control logic in Studio 5000 and designed the operator HMI from scratch in FactoryTalk View for a condensate treatment plant (activated carbon filters and mixed-bed units).
- Configured and calibrated Honeywell, Emerson and Endress+Hauser field instruments on a HART/Enraf network for control-room monitoring.
- Executed FAT and SAT validation protocols across two refinery modernization projects, travelling on-site for commissioning.
- Produced and reviewed technical documentation: P&ID, single-line and multi-line diagrams, and AutoCAD electrical and mechanical drawings; owned instrumentation, programming, commissioning and drafting end to end as one of three engineers.

Technical Support Analyst | Hikvision | Peru (Hybrid) | **August 2023 - February 2024**

- Tested new hardware and software prior to market release, identifying and reporting defects to the product team.
- Delivered 20+ technical training sessions, in-person and online, to installers and integration partners across CCTV, alarms, access control and video intercom systems.
- Specified and validated IP video infrastructure: network architecture, servers and structured cabling.

Renewals Specialist, Cisco and Fortinet | TD SYNnex | Peru (Remote) | **April 2023 - August 2023**

- Managed renewal cycles for Cisco and Fortinet service portfolios, coordinating with pre-sales engineering on technical solution scoping for enterprise clients.

EDUCATION

Graduate Certificate, Artificial Intelligence and Machine Learning (Co-op) | **December 2026** | Fanshawe College, London, ON

GPA 4.18/4.2, Dean's Honour Roll. Coursework: Deep Learning (PyTorch, TensorFlow), Data Mining, NLP, Data Visualization.

Bachelor's Degree, Electronic and Telecommunications Engineering | **2015 - 2020** | Universidad Catolica San Pablo, Arequipa, Peru

Top third of class. Specialization in Instrumentation, Control and Automation. Full scholarship to an Aerospace Engineering summer program at Shanghai Jiao Tong University (SJTU), China.

LANGUAGES

English (advanced) | Spanish (native) | French (basic)